

Practical Experiment: **Compare the response speed** **of open-source Large Language Models (LLMs)** **to find the fastest model for real-time applications**

Meeting 10 Hardware Metadata & PIP Framework Refactor

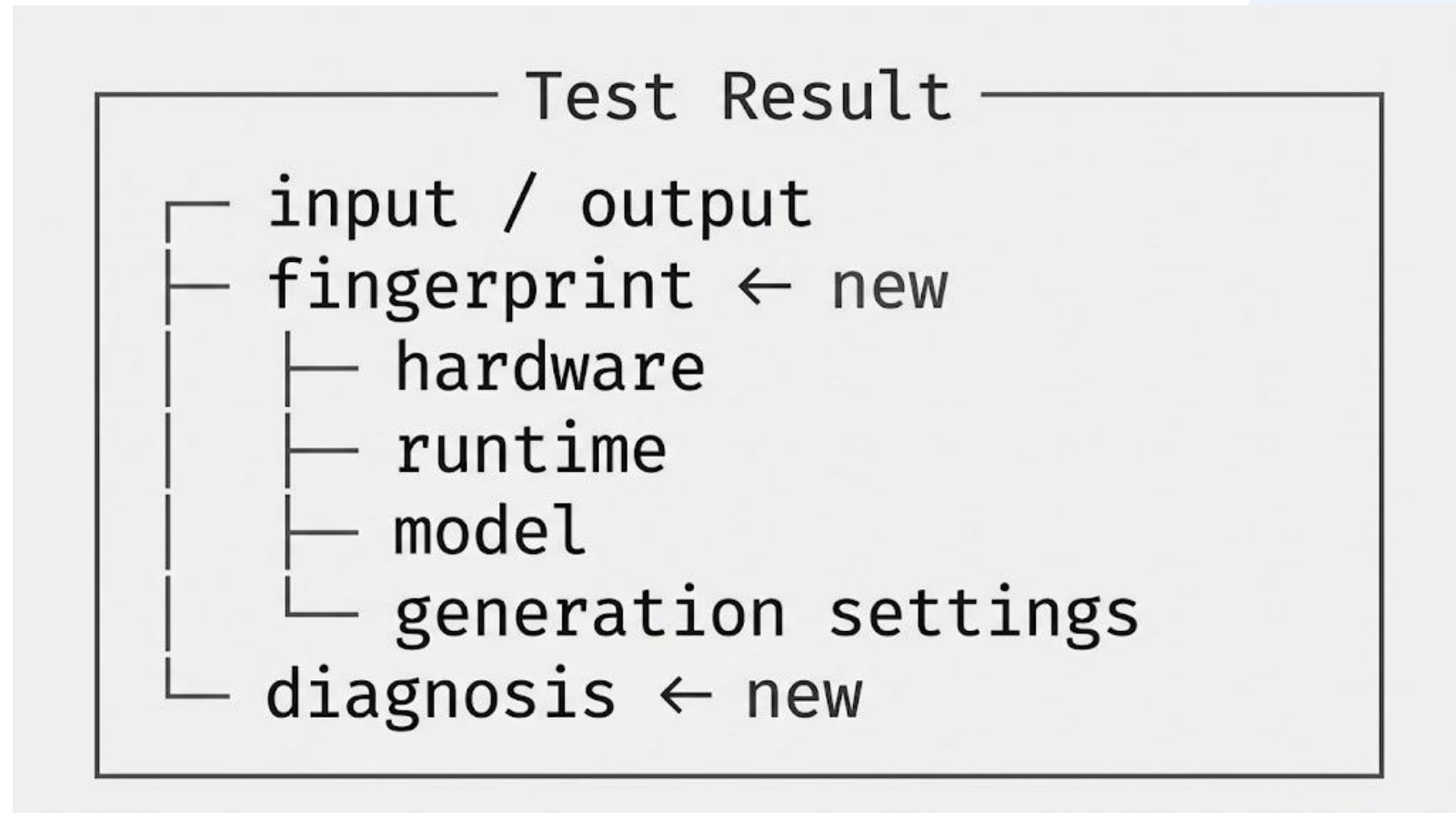
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Version: 1.0

Agenda

- Diagnosis Architecture
- Hardware & Runtime Fingerprint
- Model Details
- Generation Settings
- Diagnosis Layer
- Diagnosis Comparison



Diagnosis Architecture



Hardware & Runtime Fingerprint

Hardware Fingerprint

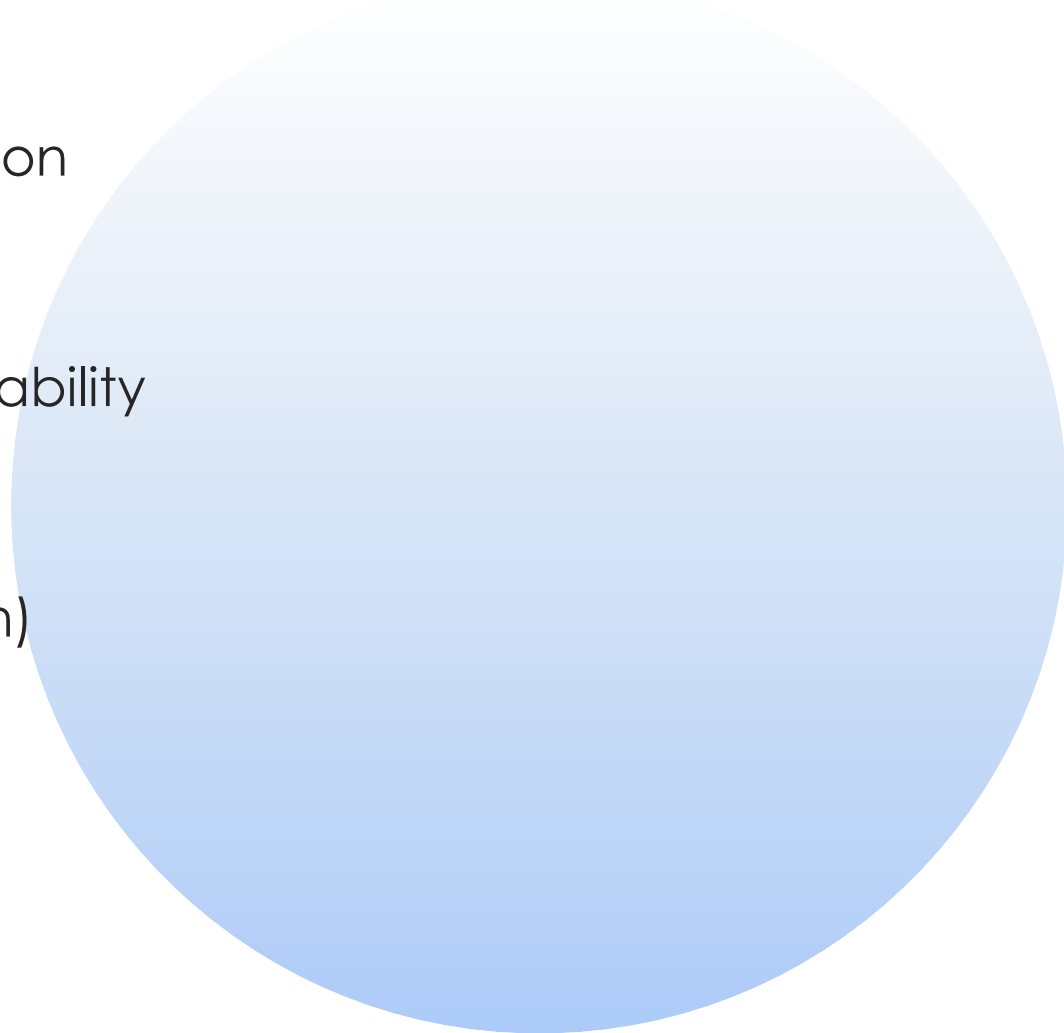
Detected on Windows and Ubuntu.

- OS, release, architecture, Python version
- CPU model and core count
- Total RAM
- GPU list with vendor and name
- Per-GPU: driver, VRAM, compute capability

Runtime Fingerprint

The engine is part of the result.

- Ollama server version (via `/api/version`)
- Ollama CLI version
- Python and library versions



Model Details



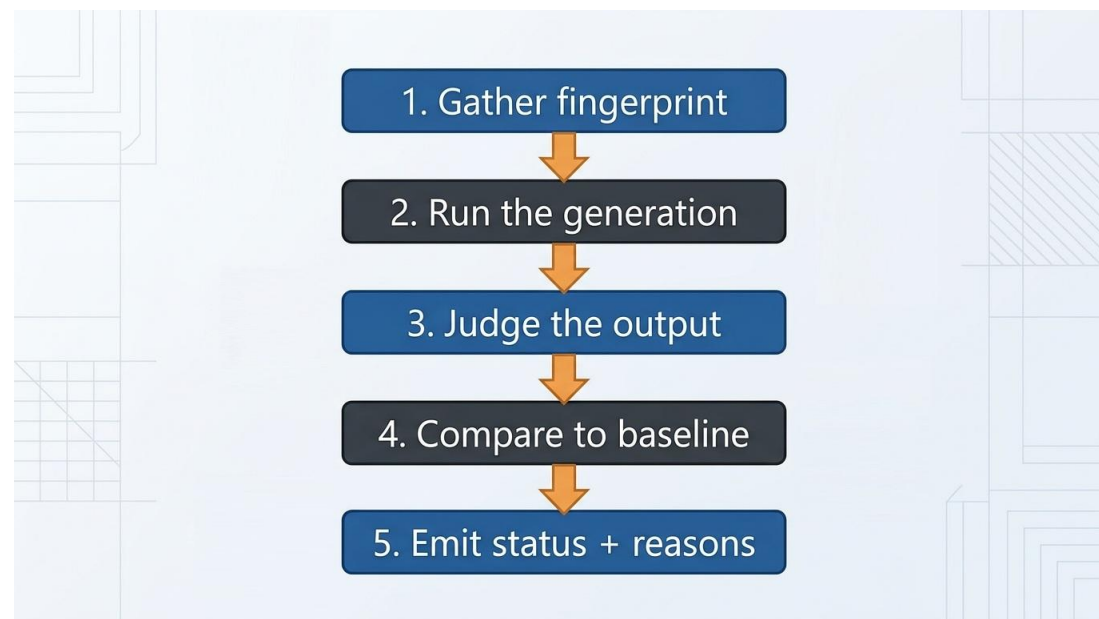
Source path	Type	Purpose
models[].digest	string	Exact artifact identity (SHA256)
models[].size	int	On-disk size
models[].modified_at	string	Last modification timestamp
models[].details.family	string	Primary family
models[].details.families	string[]	All applicable families
models[].details.parameter_size	string	Parameter count label
models[].details.quantization_level	string	Quantisation level
models[].details.format	string	Model file format

6 Generation Settings

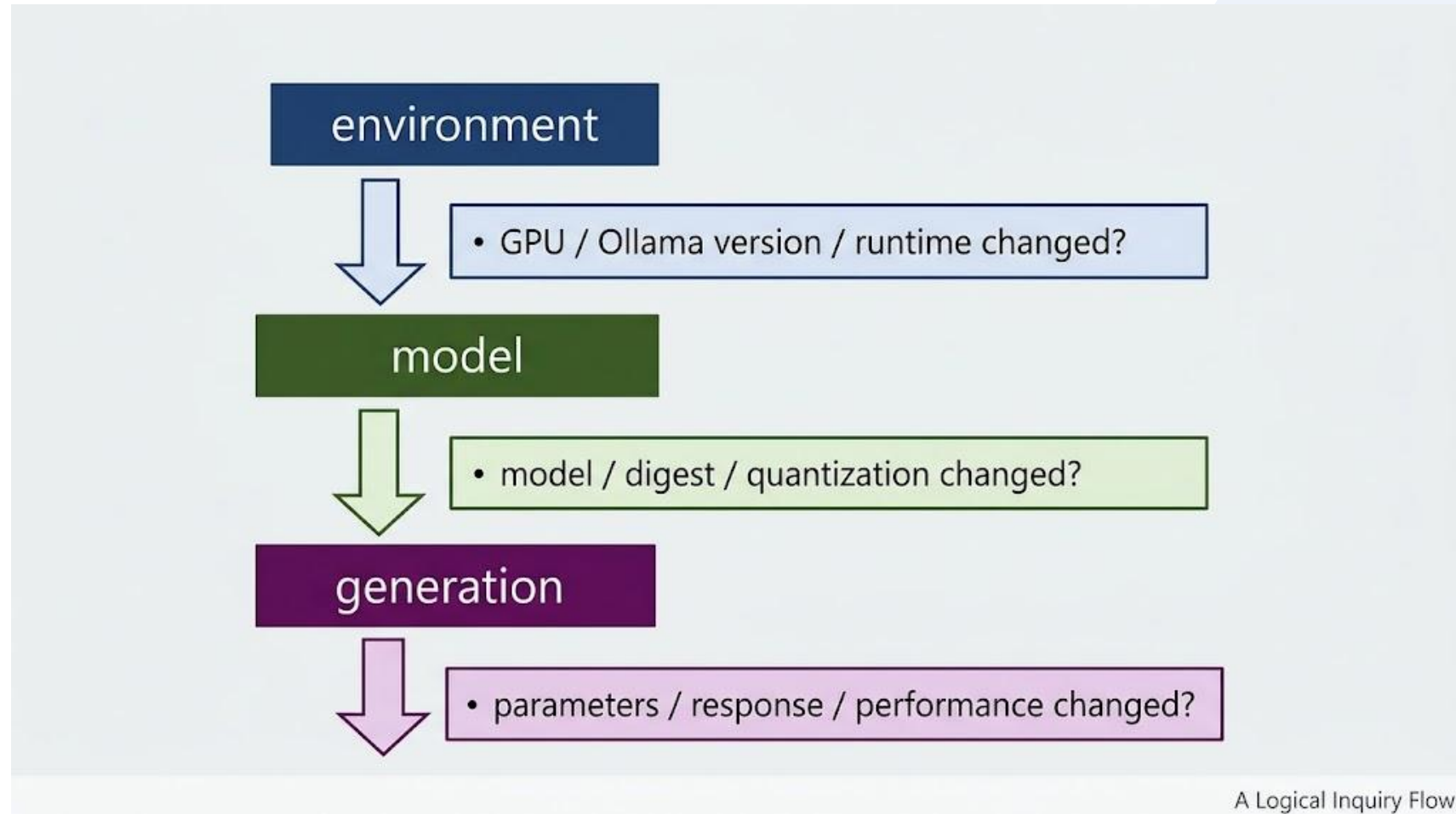
Setting	Type	Description
seed	int	Random seed for reproducible outputs.
temperature	float	Controls randomness. Higher = more creative; lower = more deterministic. Typical range 0.0–2.0.
top_k	int	Limits token selection to the K most likely tokens. Lower = more focused.
top_p	float	Nucleus sampling threshold. Only tokens within the cumulative probability mass are considered. Typically 0.9–0.95.
num_ctx	int	Context window size (tokens).
num_predict	int	Maximum number of tokens to generate.

Diagnosis Layer

Condition	Verdict
Task passed	pass
GPU set / driver / runtime changed	environment failure
Model digest / quantisation changed	model failure
Generation settings changed	model failure
Nothing changed, still failed	model failure
No baseline available	inconclusive

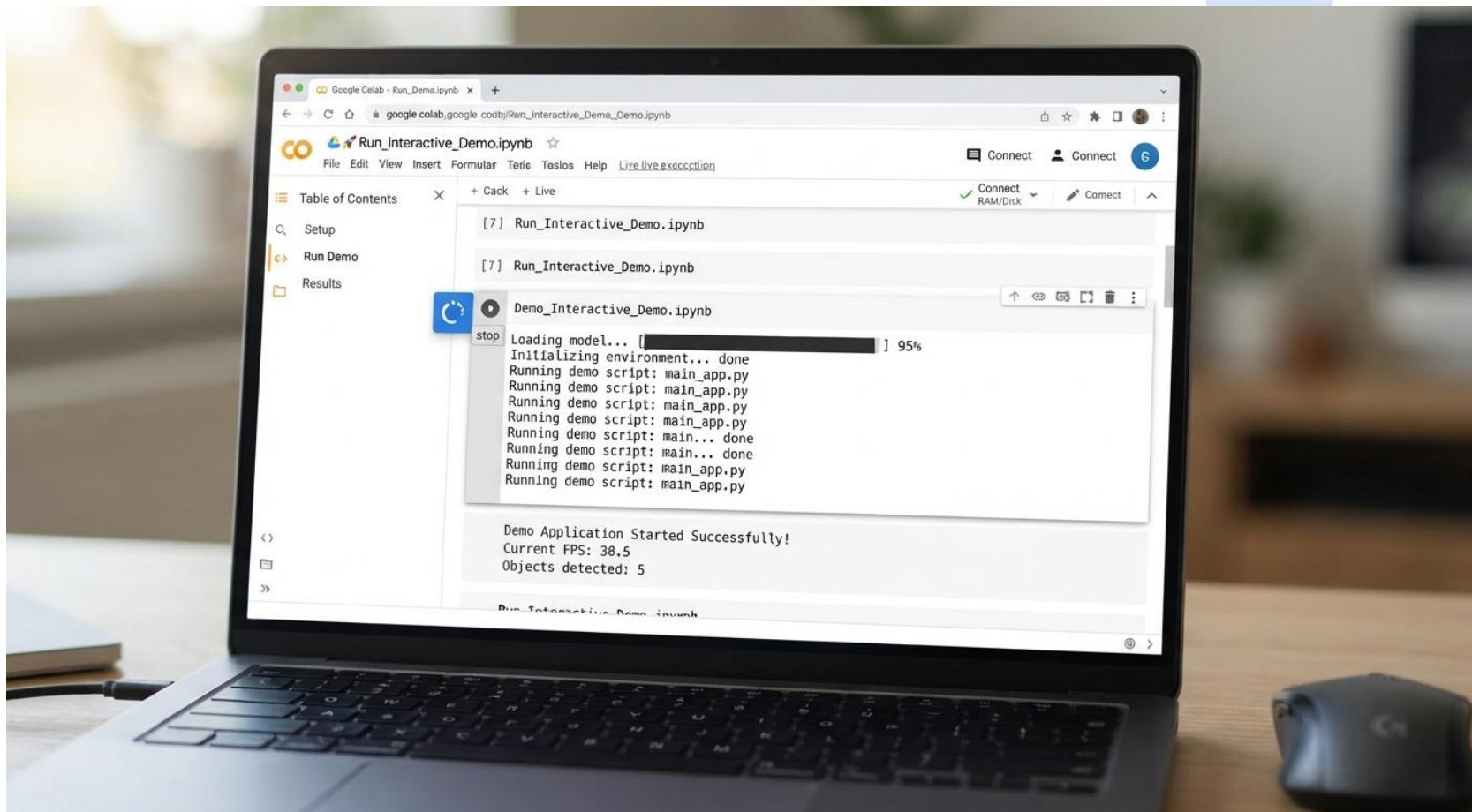


Diagnosis Comparison

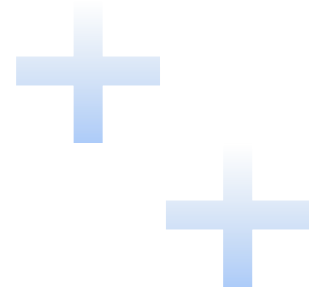


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LIVE DEMO

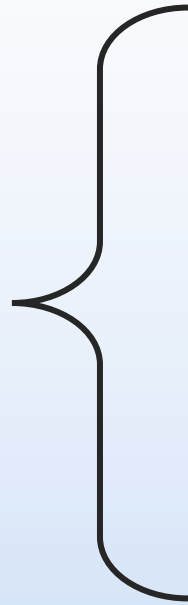


References



- Python Platform – <https://docs.python.org/3.10/library/platform.html#module-platform>
- Ollama version – <https://docs.ollama.com/api-reference/get-version>
- Ollama Model details – <https://docs.ollama.com/api/tags#response-models-items-details>
- Ollama Generation settings – <https://docs.ollama.com/api/generate#body-options>

Thank you



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